Research Statement — Reliable AI for Wireless Networks and Emerging 6G Systems

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My research centers on a practical question: How can intelligent wireless systems remain accurate and resilient when the available data, radio environment, or network operating conditions change? I earned my PhD in Electrical and Computer Engineering from the University of Oklahoma in 2025, with doctoral research focused on the application of AI/ML to wireless-network modeling, prediction, and self-optimization. My work combines communication-system knowledge with data-driven methods, including generative modeling, transfer learning, anomaly detection, and reinforcement learning.

One major contribution has been data-efficient propagation modeling and digital twins. Analytical propagation models provide a useful starting point, but they cannot fully represent the variability encountered across real environments. I studied domain-informed generative augmentation and weight transfer to improve model learning when measured radio data are scarce. This line of research investigates the quality and usefulness of synthetic data, the effect of changing domain conditions, and how predictive models generalize beyond the scenarios used in training. It led to peer-reviewed work on resilient propagation modeling and synthetic-data methods in IEEE conferences and related publications.

A second area is resilient and self-healing cellular networks. Using network measurements and simulation, I investigated intelligent outage detection, classification of abnormal operating conditions, and data-driven approaches to recovery and optimization. This work connects radio performance and network reliability with the practical constraints of autonomous management. More recently, I have explored multi-parameter optimization and proactive reliability for 5G and beyond networks. My experience with AT&T Labs, Futurewei, and AT&T has helped relate these research questions to real network measurements, diagnostic workflows, and the operational value of trustworthy predictions.

Going forward, I am interested in extending these methodological foundations toward 6G-oriented research, while recognizing that my established published work is primarily in 5G and beyond wireless AI. Promising directions include AI-native radio and network management, integrated sensing and communication, measurement-driven network digital twins, edge intelligence, and robust learning under distribution shift. I am particularly interested in how new systems can combine learning with physical constraints and uncertainty estimates rather than relying solely on black-box predictive accuracy. These interests are exploratory directions for future collaboration, not claims of completed 6G-specific projects.

At Stevens, I would welcome opportunities to collaborate on wireless communications, network intelligence, and intelligent sensing. I also see a direct connection between research and graduate instruction in EE 583 Wireless Communications: students can examine fading and interference, compare propagation models with measurements, and test model assumptions through reproducible Python or MATLAB experiments. My objective is to contribute research-informed teaching while continuing to develop methods that make emerging wireless systems more reliable, interpretable, and adaptable in practice.